

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0715

Trier 100 kV
December 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0715. The Interconnection Customer proposes a 250.9 MW wind facility interconnecting at the Trier 100 kV substation in the Indiana area. The request was received on December 15, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 238 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$487,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Imperial Substation – Chamberlin 230kV
Contingency: Hylebos – Ravenna 500kV
Loading: 121.3% (Rating: 563 MVA)
DFAX: 10.3%
Pre-project loading: 116.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Imperial Substation – Chamberlin 230kV	Rebuild line with higher-capacity conductor	\$18,577,000	3% (\$487,000)
	Total allocated to project		\$487,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.